



An empirical model of the OI FUV dayglow from DE-1 images

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Abstract

A refined empirical model of the Dynamics Explorer-1 far-ultraviolet (FUV) imaging photometer's response to Earth's quiet time FUV dayglow has been developed for thermospheric studies. The mean photometer response is based upon FUV observations in 156 images obtained during the first five months of imager operations (September 1981–January 1982) and is determined as a function of solar and satellite zenith angles, observational azimuth and solar clock angles, and solar radio flux. Variations with each parameter are characterized and, where possible, fitted with an appropriate function. The fitted response, based on the n -th power of the cosine of the solar zenith angle, is within 10% of actual mean values at all observed solar and satellite zenith angles and is consistent with the results of a first-principles calculation. Subtraction of the model background from other DE-1 images indirectly reveals the enhancement or diminution of thermospheric O/N₂ column density ratios due to transport and Joule heating effects. An analysis of summer storm-time images from the Southern Hemisphere demonstrates the use of the model in revealing these effects. The technique developed here is readily applicable to other FUV data sets. © 2000 Elsevier Science Ltd. All rights reserved.

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1. Introduction

In situ observations have shown that thermospheric composition is altered at auroral and subauroral latitudes following the onset of intense auroral activity, with greatest effects in the morning sector (Prölss, 1980, 1981, 1984; Prölss and Fricke, 1976; Prölss and Roemer, 1987). At an altitude of 280 km, for instance, increases and decreases in O density on the order of a

factor of ~ 2 are typically observed. The O/N₂ ratio, on the other hand, can decrease by more than a factor of 10, with these variations extending equatorward of 30° geomagnetic latitude during periods of intense magnetic activity. These significant variations in composition affect the brightness of Earth's FUV emissions in the sunlit hemisphere. Typically, quiet-time observations with broadband FUV instruments reveal a steady increase in brightness with decreasing solar zenith angle and brightening near the dayside limb at all solar zenith angles. Notable exceptions are observed after the onset of intense auroral activity, when large ($\sim 50\%$) decreases in the brightness of OI emissions at

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130.4 and 135.6 nm are observed in the morning local time sector at auroral and subauroral latitudes (e.g., Craven et al., 1994; Meier et al., 1995; Nicholas et al., 1997).

The association between the changes in composition and FUV brightness has been studied in detail and can be exploited for remote determination of thermospheric composition. Both broadband and spectrally separated observations of the terrestrial FUV emissions have been used to infer thermospheric composition variations from changes in Earth's FUV signature (e.g., Conway et al., 1988; Meier, 1970; Meier and Prinz, 1971; Parish et al., 1994; Strickland and Thomas, 1976). The development of sophisticated models has been necessary to correctly deduce composition changes from FUV observations (Gladstone, 1994; Link et al., 1988; Meier and Lee, 1982; Strickland et al., 1994; 1995). Recent work by Strickland et al. (1998) has demonstrated the use of FUV modeling techniques in determining O/N_2 ratios over the full field-of-view of the DE-1 FUV imager. In many of these works, there is a need to establish the reference quiet-time FUV brightness.

The objectives of this report are (1) to study quiet-time FUV variations and their relation to composition, and (2) to develop an empirical model as a comparative reference for detecting changes in composition due to magnetic activity or other influences. This is done with the requirements that the model reflect all possible observational geometries (Sun-Earth-satellite positions) and that it account for (1) changes in solar inputs and instrument sensitivity occurring over the several years of DE-1 observations, (2) normal variations in quiet-time thermospheric composition on the day-side, and (3) variations in OI emission profiles with solar zenith angle.

A detailed quantitative analysis of DE-1 FUV images by Meier et al. (1995) has firmly established the relative contributions to the photometer's response using filter #2: 85–90% (OI 130.4 nm); 5–8% (OI 135.6 nm) and 5–8% (N_2 LBH). For a given observational geometry, modeling work by Strickland et al. (1998) demonstrates that the relationship between photometer response and O/N_2 in an atmospheric column of a particular N_2 column density is nearly linear, with an uncertainty of $\sim 10\%$. Presented here is an empirical determination of the average quiet-time response of the photometer from actual observations, deviations from which are directly related to changes in O/N_2 . Previous analyses have shown variations in OI image brightness as percent differences (PD) from reference quiet-time values (Craven et al., 1994; Immel et al., 1997a; Nicholas et al., 1997). The ability to convert PD to O/N_2 requires that the empirically-derived quiet-time dayglow values accurately model quiet-time FUV brightness for all observational geometries and

levels of solar activity. Hence, it is important that the reference values be well established and that the model account for all possible variations in quiet-time brightness appearing in individual images.

The earliest generation of this empirical quiet-time model (Nicholas et al., 1997) makes no corrections for azimuth of observation or solar UV fluxes. The Nicholas et al. method has recently been utilized by Strickland et al. (1998) to generate reference images for comparison to a first-principles numerical model. The current empirical model provides substantial improvements, which are described in this report.

2. Modeling method

2.1. Image selection and initial processing

The empirical model is based on multiple observations (i.e., image pixels) of the quiet-time FUV dayglow by the DE-1 Spin-Scan Auroral Imager (Frank et al., 1981) from which the average photometer response, $\langle r \rangle$, is calculated. Images are selected with the requirements that (1) values of the AE index must remain below 100 nT for a minimum of 6 h, (2) solar flare activity not be high during the 6 h prior to imaging, and (3) there must be a favorable imaging geometry. Observations were limited to the period 23 September 1981 (first day of imaging operations) through 19 February 1982 (last day of operations before the first Sun crossing of the imaging field-of-view). An extension to later time intervals will require consideration of systematic decreases in the photometer's sensitivity across the pass band due to degradation of FUV materials (e.g., lens and mirror coatings) (Rairden et al., 1986).

A total of 185 images from 16 days meet the initial selection criteria. Several of these images are strikingly unrepresentative of the remainder of the quiet-time set, with significant deviations of measured brightness from average values. These 29 images from three separate days have been removed from the quiet-time set, so the final number of images used in the quiet-time model is 156. Images from one day in October, five days in November, three days in December, and four days in January are used.

Certain regions of each image are excluded from initial modeling efforts as brightness variations in these regions are not directly related to variations in thermospheric composition. One such region is Earth's limb, where the contribution of N_2 LBH band emissions to the total photometer response (using filter #2) increases strongly since the thermosphere is optically thin to these emissions. Another is the area of high magnetic latitudes, where auroral emissions are comparable in

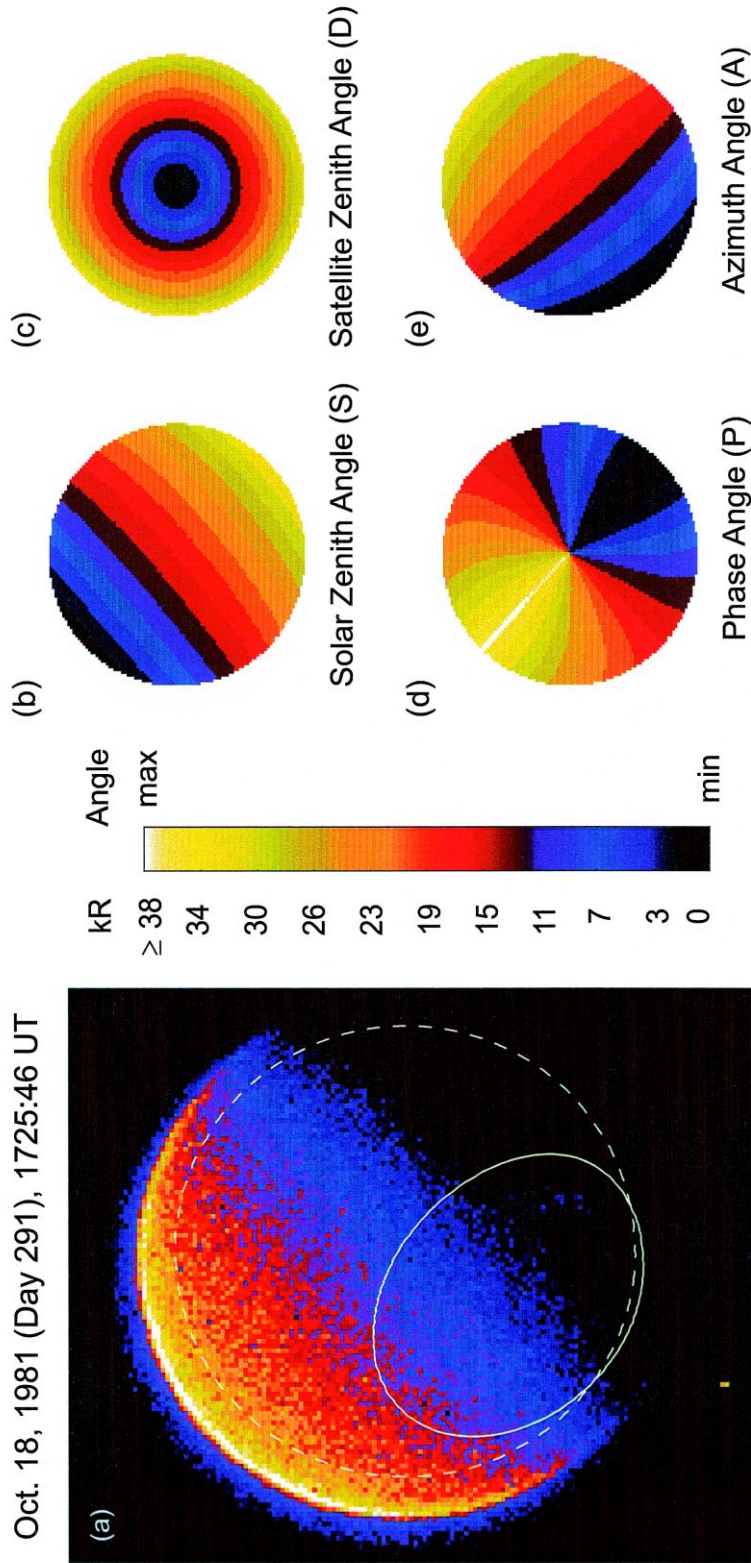


Fig. 1. DE-I FUV image and geometric angles for modeling. (a) Original DE-I image of Earth at 17:25 UT on 18 October 1981 (day 291), at wavelengths 123–165 nm. The Sun is toward the upper left in this full image obtained from an altitude of $3.44 R_e$. Brightness in kilorayleighs, kR, is coded to the right of the image. The color bar indicates both brightness and the range of angles, where the corresponding minimum and maximum angular values are reported here. The pertinent angles for this image are shown to the right. (b) Solar zenith angles, S , ranging from 4 to 151° . (c) Satellite zenith angles, D , ranging from 0 to 180° . (d) Azimuth angles, A , ranging from 0 to 180° . (e) Phase angles, P , ranging from 30 to 188° .

FUV brightness with the local dayglow and represent a “contamination” of the dayglow. To exclude these effects, each image has been divided into limb, disk and auroral regions, as illustrated in Fig. 1(a) with a quiet-time image from 18 October 1981. The dashed outer contour indicates the limb-disk separation angle, empirically determined by Nicholas (1993). The low-latitude limit for the quiet auroral region is defined as 65° geomagnetic latitude, shown by the inner solid contour in Fig. 1(a), wherein faint auroral emissions are marginally visible. Only samples (image pixels) from the disk region which are not in the auroral region are given further consideration in constructing model values.

2.2. Parameters

The empirical model of photometer response is based strictly on imaging geometry and solar radio flux. Imaging geometry is specified using four angles. The solar zenith angle (S) and DE-1 observation angle (D) are those between the zenith (at the location of a particular pixel) and a vector towards the Sun and satellite, respectively. These two angles are the main parameters by which the photometer’s response is quantified. The variation in photometer response with S is the dominant effect, as the response increases by $\sim 300\%$ between $S = 80^\circ$ and 20° , for example. The maximum variation for $D = 10\text{--}60^\circ$ is on the order of $\sim 10\%$, a variation rooted in the increase with look angle of the length of the optical column of emissions to which the thermosphere is optically thin (e.g., OI 135.6 nm and N_2 LBH). These are the primary variations which must be removed from images to reveal effects of magnetic activity. The variation in these angles for the quiet-time image of Fig. 1(a) are shown in Figs. 1(b) (S) and (c) (D). Values of S range from 4° to 151° and D ranges from 0° to 90° . The model by Nicholas et al. (1997) is based on these two parameters alone. The observational azimuth angle, A , is traditionally that between the projection of the Sun and viewing (satellite) vectors onto the local horizon. Azimuth angles calculated for the image shown in Fig. 1(a) are shown in Fig. 1(d), where the full range from $0\text{--}180^\circ$ can be seen. The phase angle, P , is basically a clock angle about the sub-solar point. It is lowest in the afternoon sector, with values ranging from 30 to 188° present in the image as shown in Fig. 1(e). The importance of A and P is described below.

The sensitivity of the photometer’s response to azimuth angle, A , (at fixed S and D) is introduced into this analysis of DE-1 images by the assumption of a single, fixed emission altitude and is also inherent to observations near the solar terminator. The OI 130.4-nm emission source function is actually significant over

a large range of altitudes (100–500 km) and its height profile changes with S and solar activity (Meier, 1991). At any S where the selected altitude does not correspond to the actual altitude of the peak in the emission source, an apparent variation in brightness with A will result. For example, if the selected altitude is higher than the actual peak emission altitude, observations will look beyond the point of interest to brighter ($A > 90^\circ$) or dimmer ($A < 90^\circ$) regions of the FUV dayglow. A second-order effect is that even if the peak emission source altitude matches the assumed altitude, variations in A may cause the line of sight to pass through a higher altitude region of variable brightness. This is especially important at the terminator where the OI 130.4-nm source function changes most rapidly with S . We select 500 km as the altitude at which to calculate all angles for the empirical model, with the assumption that these azimuthal variations can be quantified and corrected. The selection of this particular altitude is discussed in the Appendix.

Two additional parameters are included in the modeling technique to better quantify changes in Earth’s FUV signature due to quiet-time variations in thermospheric composition and changes in solar input. The phase angle, P , is defined in an orthogonal right-handed coordinate system where $\hat{x}$ is in the direction of the Sun from Earth, $\hat{z}$ is northward in a plane containing $\hat{x}$ and Earth’s rotational axis and $\hat{y} = \hat{z} \times \hat{x}$. P is measured in the y – z plane and is 0° along the y -axis in the afternoon sector. Together, P and S form an orthogonal pair of angular coordinates for specification of points on the disk with respect to the Sun. Diurnal variations in heating of the thermosphere will cause changes in O/ N_2 column densities, and consequently OI FUV brightness, with local time. S and P are combined to quantify these variations and provide corrections such that quiet-time changes in FUV brightness across the disk may be distinguished from geomagnetic effects. The effect of diurnal heating on Earth’s FUV signature is evident in other modeling efforts. OI brightness variations can be seen at high, fixed values of S in simulations of DE-1 images generated by Meier et al. (1995). These simulated images clearly show that contours of constant 130.4-nm OI brightness do not follow contours of equal S . The inclusion of P is thus required in a geometrical model of DE-1 FUV photometer response.

Important parameters for determining the overall brightness of Earth’s FUV dayglow are the solar EUV and FUV output. Solar radio flux, F , is used as a proxy for solar EUV and FUV fluxes, as measurements of the full solar UV spectrum were not made throughout the time of the quiet-time observations. A known correlation between solar UV intensities and F justifies its use in characterizing the brightness of the FUV dayglow (Barth et al., 1990; Hedin, 1984),

though the correspondence is best on time scales greater than several days. It is expected therefore that the photometer's response for a given observational geometry will bear some relation to the solar radio output.

Two instrumental parameters may influence the photometer's response during the study interval: (1) the dependence on photometer response on stepping mirror angle and; (2) the decrease in instrument sensitivity with time. The possibility of variation in photometer response with stepping mirror angle (and thus, the intersection of the photometer line-of-sight with the primary mirror) was discussed by Rairden (1985). An asymmetry in the reflectivity of the optics to HI Ly α emissions on the order of a factor of two between minimum and maximum excursions of the stepping mirror exists. However, our analysis of photometer response to OI emissions shows that the sensitivity varies significantly over only 1/3 of the stepping positions (i.e., scan lines), with a gradual $\sim 20\%$ reduction in sensitivity after scan line 80 (of 120) near maximum mirror excursion. With the limb region removed and the local time of the satellite orbit proceeding toward early morning hours, the effect is that very few of these pixels are included in the model; 7% of all individual measurements (pixels) within the disk region of quiet-time images were in this range of scan lines. These pixels are excluded from the calculation of $\langle r \rangle$ except for pixels at $S > 80^\circ$ and $D > 50^\circ$, where these measurements are required for complete specification of the photometer response.

Variation in photometer sensitivity is assumed to be negligible in the time during which quiet-time images were obtained (September 1981–February 1982). For analysis of images obtained later in the course of the mission (e.g., February 1983) it is necessary to make corrections for the reduced sensitivity of the instrument. The sensitivity does not decrease evenly across the passband, and the relative contribution of the optically 'thin' OI 135.6-nm and N $_2$ LBH emissions to the total photometer response are greater in 1983 as evidenced in the greater dependence on D . An accurate correction that accounts for both of these factors is achieved by comparing the 1981 quiet-time model to the photometer response in 31 images obtained during periods of low magnetic activity on 2 and 11 January 1983. The ratio of these average photometer responses is fitted with a function varying linearly in S and D . The coefficients of this fit are given in the Appendix.

2.3. Technique

The modeling technique involves first binning the quiet-time photometer response, r , from the 156 images in terms of S and D and each of the three remaining

parameters individually (A , P , or F). It has been determined but not shown here that the influence of P and F on r does not vary significantly with D . For this analysis, therefore, these parameters are characterized as a function of S alone. The dependence of r on A at the nadir ($D = 0^\circ$) is identically zero, but becomes relatively large near the limb and terminator. The variation with this angle is therefore determined as a function of both S and D .

In the second step, an iterative technique is used to separate the effects of each parameter from one another and determine final corrections to the average response, $\langle r \rangle$. An example of a possible source of aliasing that is minimized by this method is the

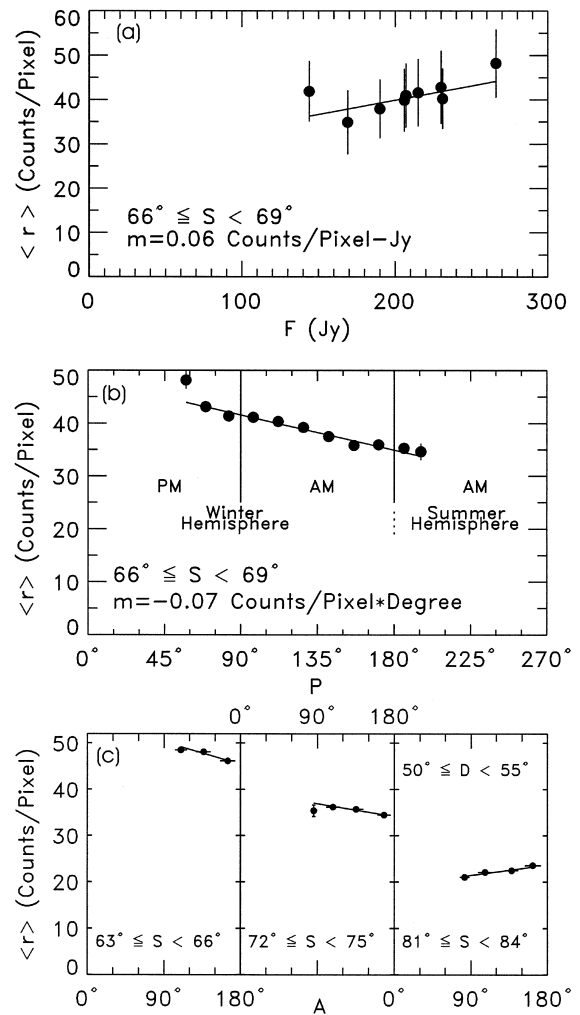


Fig. 2. Initial fits to variation of $\langle r \rangle$ with angular parameters and solar flux. (a) Fit to slope of $\langle r \rangle$ vs F at $66^\circ \leq S < 69^\circ$. (b) Fit to slope of $\langle r \rangle$ vs P at $66^\circ \leq S < 69^\circ$. (c) Slope of $\langle r \rangle$ vs A at $50^\circ \leq D < 55^\circ$ and three increasing ranges of S .

change in F as DE-1 observations tended toward earlier local times (increasing P). It is clear that some sources of FUV variations are not completely independent (e.g., variations in F will affect the OI 130.4-nm source function and therefore the dependence of $\langle r \rangle$ on A), but the data from 156 images provides insufficient coverage to fully characterize all the variations in dayglow brightness simultaneously. Furthermore, variations in FUV brightness with A and P are on the order of only 10% for most solar zenith angles. A variation in the response that depends on the product $A \cdot P$, for example, would presumably be even less significant over the sunlit disk.

The iterative process begins with the determination of the dependence of $\langle r \rangle$ on F at all values of S (in 3° bins). A linear function is least-squares fitted to the variation and all quiet-time images are corrected to $F = 200$ Jy. An example of a fit for $66^\circ \leq S < 69^\circ$ is shown in Fig. 2(a), where the expected increase in $\langle r \rangle$ with F is apparent. The slopes determined here provide a means to correct all quiet-time images to $F = 200$ Jy. Dependence of the corrected average response, $\langle r(F = 200 \text{ Jy}) \rangle$, on P is then determined at all values of S (again using 3° bins of S). That variation is least-squares fitted as well, an example of which is shown in Fig. 2(b) for $66^\circ \leq S < 69^\circ$. Note the clear variation of $\langle r(F = 200 \text{ Jy}) \rangle$ with P , where the brightness increases toward the afternoon sector. Two AM sectors are indicated, one in summer, where $P > 180^\circ$. From the slope of these fits, one can normalize quiet-time images to fixed values of P . These corrections are applied and the dependence of $\langle r(F = 200 \text{ Jy}, P = 90^\circ) \rangle$ on A is found as a function of both S and D ($\Delta S = 3^\circ$, $\Delta D = 5^\circ$). The variation with A is shown at three different ranges of S at $50^\circ \leq D < 55^\circ$ in Fig. 2(c). In this range of D , the fit to $\langle r(F = 200 \text{ Jy}, P = 90^\circ) \rangle$ as a function of A has a slope which changes from negative to positive values between $S = 72^\circ$ and $S = 81^\circ$.

At the outset of the second iteration, the first iteration corrections for A and P alone are applied to quiet-time images in order to determine an A - and P -corrected F dependence that is superior to that obtained in the first iteration. This is repeated until each dependence changes insignificantly between iterations; i.e., values of i -th iteration are all within the range of uncertainty of values from the previous iteration. The analysis shows that the A and P dependencies determined in the third iteration do not differ significantly from the results of the second iteration anywhere in the observed range of S . The F dependence determined in the fourth iteration is similar to that found in the third. The variations determined in the final iteration are discussed in the Appendix, where fits to the variation in $\langle r \rangle$ with each parameter are pre-

sented. The physical origin of each variation is also discussed.

2.4. Final S and D dependence

The mean photometer response $\langle r \rangle$ is determined as a function of S and D only after scaling the value of each disk-region pixel to $A = 90^\circ$, $P = 90^\circ$ and $F = 200$ Jy. The corrected images are binned in increments $\Delta S = 1^\circ$ and $\Delta D = 6^\circ$. The smaller bin size for S is needed to characterize the more rapid variations in dayglow brightness with S , as compared to the weak linear dependence on D . The mean photometer response $\langle r \rangle$ is then computed for the contents of each bin. Values of $\langle r \rangle$ thus obtained are summarized in the projection of Fig. 3(a) as functions of S and D . Note that there exist ranges of S and D values where no measurements were obtained, particularly at low values of S .

It is useful to select a well-behaved function to represent the values of $\langle r \rangle$ so that it can provide a reliable extrapolation of the photometer's response at values of S and D not available in the original quiet-time data set, but which do exist in other images obtained in the mission (e.g., February 1983). For example, a least-squares fit to the data with a polynomial function dependent on S and D results in fitted values that closely match the data over most of the range of S and D sampled (Nicholas et al., 1997). However, such a function rapidly departs from a realistic representation of the photometer response at S and D outside the domain of the quiet-time observations and at the terminator. This is demonstrated by Strickland et al. (1998), who compare the fit by Nicholas et al. to first-principles modeling results for selected images and show a poor correspondence between the models. The use of a trigonometric function of the form $\langle r \rangle = B \cdot \cos(S)^n$ results in a much improved fit to the data at fixed D and is well-behaved at low values of S . Through several trials it has been found that this function provides a good fit to the data for $S \leq 80^\circ$. A separate treatment is required for greater S , as an attempt to fit the trigonometric function for greater zenith angles results in increasingly large deviations of the fit from the data at $S > 60^\circ$.

The trigonometric fit to $\ln \langle r \rangle$ is carried out for $S \leq 80^\circ$ as a function of $\cos(S)$ in each 6° bin of D , resulting in coefficients $\ln(B)$ and n for each bin of D . A linear function is then separately fitted (least-squares, no weighting) to all the values of $\ln B$ and n so that values of these coefficients that vary smoothly over the range of D are found. For the function

$$\langle r \rangle = B \cdot \cos(S)^n \text{ counts/pixel} \quad (1)$$

the coefficients that provide the best fit to $\langle r \rangle$ are

$$\ln B = ((4.32 \pm 0.04) + (0.002 \pm 0.001)$$

$$\cdot D) \ln(\text{counts/pixel}) \quad (2)$$

$$n = (0.75 \pm 0.04) - (0.003 \pm 0.001) \cdot D. \quad (3)$$

Given D , this provides for the calculation of B and n , and subsequently $\langle r \rangle$, at any S . The variation of each term with D is small but significant.

No fit is performed for $80^\circ < S \leq 105^\circ$, as an appropriate function was not determined. Instead a table of values of $\langle r \rangle$ is constructed in $\Delta S = 1^\circ$, $\Delta D = 1^\circ$ increments. A $3^\circ \times 3^\circ$ boxcar averaging technique is applied to smooth the data after bins with no samples are replaced with interpolated values. These data are rebinned to $\Delta S = 3^\circ$, $\Delta D = 5^\circ$ and reported in tabular form in the Appendix (Table A2). Using these data instead of a fitted function eliminates the large percent deviations that occur when a function such as a quadratic is fit to data that range from 10 counts ($S = 80^\circ$) to fractional counts ($S = 105^\circ$). Such a function tends to show large percent deviations from the fitted data at low counts.

A 3-D projection of the reference values $\langle r \rangle'$ calculated from these two methods is shown in Fig.

3(b) for $S = 0-105^\circ$ and $D = 0-65^\circ$. The trigonometric fit and smoothed tabular data meet at $S = 80^\circ$. A final $5^\circ \times 5^\circ$ boxcar averaging is performed in the region $S = 75-85^\circ$ for all D . This smooths slight discontinuities between the trigonometric function and smoothed data values. The projection of Fig. 3(b) is in the same format as that shown by Strickland et al. (1998, Figs. 1 and 4(a)) for quiet-time reference response, though with greater extent in S (105 vs 90°) and lesser extent in D (65 vs 90°).

Residual percent differences between the fitted and original data, of Fig. 3(b) and (a) respectively, are shown at four different bins of D in Fig. 4. Uncertainties in the PD values are indicated at every third bin of S with a vertical bar, and the vertical dashed line at 80° indicates the separation between values of $\langle r \rangle'$ obtained from the $\cos(S)$ function and those found in Table A2. There are clear trends in the PD values, due to the inability of the $\cos(S)$ function to perfectly duplicate the dayglow variation. However, it is clear that the reference values $\langle r \rangle'$ represent the measured response of the photometer to within $\sim 10\%$ for all D at almost all $S < 95-100^\circ$, as no larger percentage deviations are present.

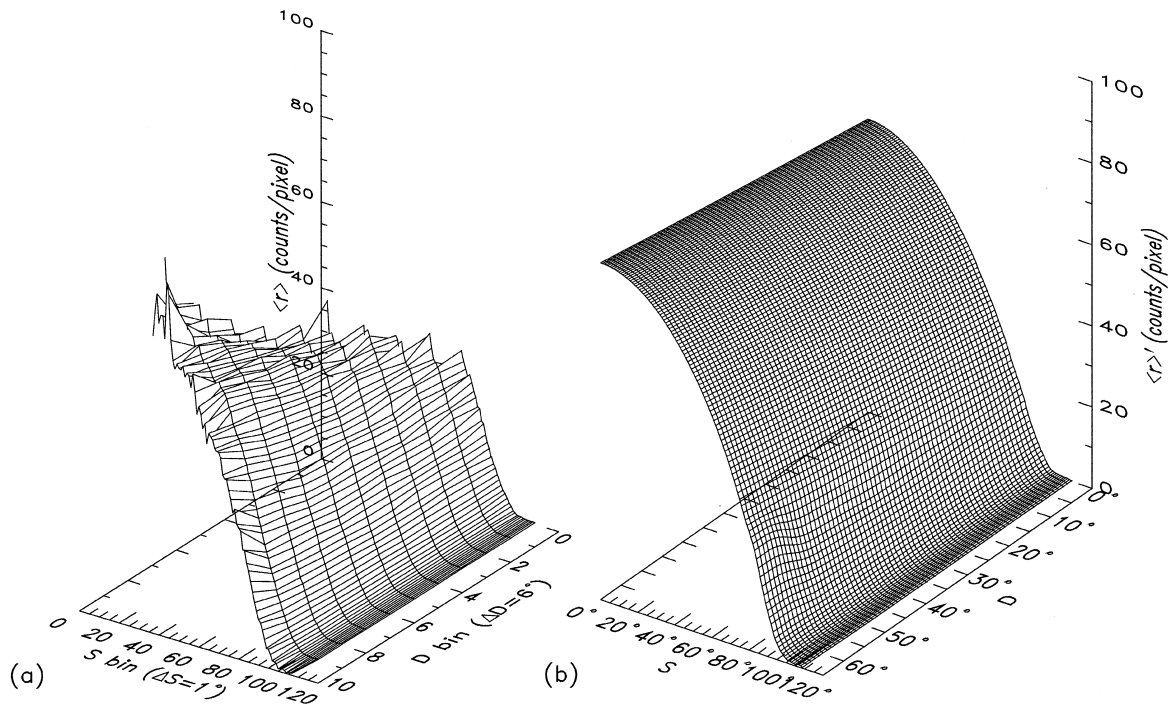


Fig. 3. (a) Average response $\langle r \rangle$ of the DE-1 FUV photometer for the quiet-time data set for filter #2 (123–165 nm) for zenith angles $S = 0-105^\circ$ and $D = 0-65^\circ$, with $\Delta S = 3^\circ$ and $\Delta D = 5^\circ$. The response in individual images is normalized to $A = 90^\circ$, $P = 90^\circ$ and $F = 200$ Jy. (b) Calculated reference values $\langle r \rangle'$ for the normalized DE-1 FUV photometer response of Fig. 3(a). The solar and satellite zenith angles are $S = 0-105^\circ$ and $D = 0-65^\circ$ respectively, and are now shown in 1° bins.

3. Validation

3.1. Comparison to first-principles model

The empirically determined model values can be compared to first-principles modeling results to test the validity of this technique and the $\cos(S)^n$ based fit to $\langle r \rangle$. Strickland et al. (1998) have calculated the predicted DE-1 photometer response for particular S and D and O/N₂ column density ratio using radiative transfer techniques and the MSIS model atmosphere. A direct comparison to these predictions is made in this section. Average column densities corresponding to each value of $\langle r \rangle$ of Fig. 3(a) are also calculated for this comparison. This is done by replacing the response of the individual pixels in the 156 images of the quiet-time set with O and N₂ column density values (determined using MSIS and the formulation described in Section A2 of the Appendix) and then determining $\langle O \rangle$ and $\langle N_2 \rangle$ column densities and their ratio as a function of S and D . This provides an estimate of atmospheric conditions present during the quiet-time observations discussed in Section 2 of this report, and though

obtained in a similar manner, is entirely separate from the determination of $\langle r \rangle$.

Figs. 1 and 4(a) of Strickland et al. (1998) show the predicted response at particular values of S and D , where $0^\circ < S < 90^\circ$ and $0^\circ < D < 90^\circ$. Empirical model values (shown in Fig. 3(b) of the current work) are available for comparison in this entire range of S and for $0^\circ < D < 65^\circ$. The first-principles predictions are for a fixed thermospheric O/N₂ column density ratio of 0.98 and represent solar conditions on 24 September 1981 (Strickland, private communications), at which time $F = 182.2$ Jy. The fixed column density ratio is achieved by determining the altitude of the base of the N₂ column with a density of 10^{17} cm⁻² and scaling the O number density profile in the same altitude range. In contrast to the first principles work, the average O/N₂ column density ratio in the 156 quiet-time images is generally between 0.95 and 0.70, with lowest values at high S . No correction is made to account for this difference, though the empirical model is corrected to $F = 182.2$ Jy (from 200.0 Jy) using the correction factors listed in Section A1 of the Appendix.

The values of P and A for the first-principles model are of some concern, but P tracks variations in O/N₂ which do not apply for the case of fixed O/N₂. Azimuth angle, A , has greatest influence on photometer response in the disk-region in the $80^\circ < S < 105^\circ$ range. The first-principles work does not include this angle and partially avoids the resulting increase in uncertainty by not attempting to predict photometer response at $S > 90^\circ$.

Empirically determined and first-principles values of the average response (denoted by $\langle r \rangle'$ and $\langle r \rangle_{FP}$, respectively) are directly compared at selected points of S and D in the surface plot of Fig. 5(a). The 3-D projection of these $\langle r \rangle$ values is similar to that of Fig. 3(b), and the values of $\langle r \rangle_{FP}$ are exactly those from Figs. 1 and 4(a) of Strickland et al. (1998). Heavy lines indicate $\langle r \rangle'$ and thin lines denote $\langle r \rangle_{FP}$. This initial comparison demonstrates the general agreement between the models, though the empirically determined values are greater by ~5–15% at most S and D , except near the terminator, where some $\langle r \rangle'$ values are marginally lower than $\langle r \rangle_{FP}$. The relationship between the two parameters is examined more closely in Fig. 5(b), where percentage difference values (PD) between the two parameters are shown, for example, at $0^\circ < S < 90^\circ$ and $D = 58^\circ$. The PD values slowly increase from 10 to 15% with S in the $S = 0$ – 75° range, then drop rapidly to 2% at $S = 90^\circ$.

The average quiet-time O/N₂ column density ratio, $\langle O/N_2 \rangle$, is shown at the same S and D in Fig. 5(c), where the fixed value of 0.98 used for calculation of $\langle r \rangle_{FP}$ is indicated with a dashed line. These data do not extend to $S = 0^\circ$ as no quiet-time observations of

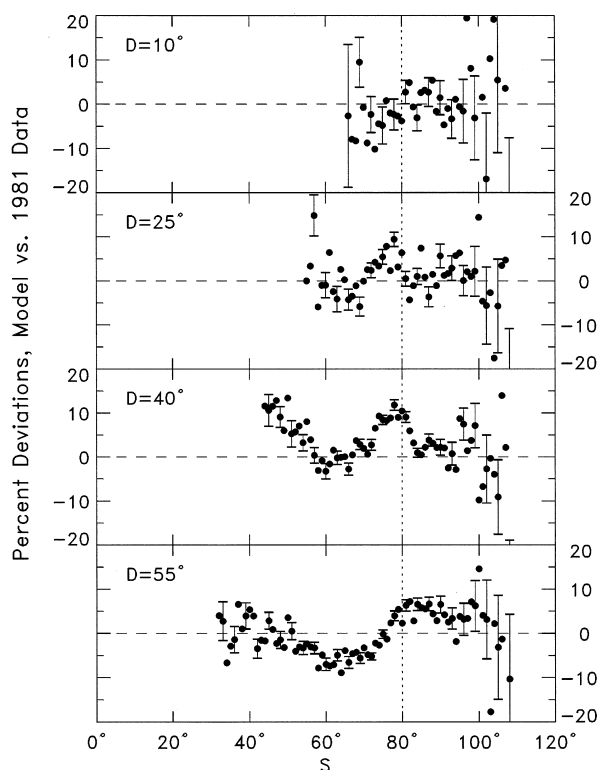


Fig. 4. Percentage deviations of the average response $\langle r \rangle$ (Fig. 3(a)) from the reference values $\langle r \rangle'$ (Fig. 3(b)) for all S and four values of D : (a) $D = 10^\circ$, (b) $D = 25^\circ$, (c) $D = 40^\circ$ and (d) $D = 55^\circ$.

Earth's disk contained pixels at these angles (the range of coverage in S and D can be seen in Fig. 3(a)). The MSIS-determined values of $\langle O/N_2 \rangle$ increase from ~ 0.8 to 0.9 with increasing S to $S = 65^\circ$, then drop to 0.7 as S increases from 65° to 90° . This corresponds reasonably well to the variation in PD in this range of S and D . With this decrease in $\langle O/N_2 \rangle$ one expects that the average response, $\langle r \rangle$, must also diminish in comparison to a model where the O/N_2 ratios are fixed. Indeed, the PD values show this to be the case.

Though the variation of PD with $\langle O/N_2 \rangle$ demonstrates the expected trend, the result that $\langle r \rangle'$ is greater than $\langle r \rangle_{FP}$ at most S is surprising, since the column ratio for the first-principles work is fixed at $O/N_2 = 0.98$, a value greater than most observations in the quiet-time set. The different techniques used in calculating column densities may play a role: the N_2 column density above the 10^{-5} Torr base pressure-altitude is 2–3 times larger here than the fixed value of 10^{17} cm^{-2} used by Strickland et al. (1998). Given the difference in scale heights of O and N_2 , the additional column length at the base may be responsible for reducing the ratio to values well below unity. Values above unity are necessary to provide an explanation for the observation that the observed dayglow brightness is

greater than that which Strickland et al. predict for $O/N_2 = 0.98$.

3.2. Application to quiet-time images

Images obtained during a period of low magnetic activity are presented in this section to demonstrate the ability of the model to represent quiet-time FUV brightness. The low level of magnetic activity on 31 October 1981, is evident in the auroral electrojet indices (AE) shown in the left-hand panel of Fig. 6. Sixteen images obtained between 05:52 and 09:06 UT of day 304, 1981, are analyzed here, a period indicated with vertical lines in the AE plot. PD values for each disk-region pixel of the sixteen images are calculated by first determining the difference between the observed response and the A , P and F corrected $\langle r \rangle'$ value calculated given the S and D values for the image pixel, and then dividing the difference by the same corrected model value. PD values are determined for all pixels in each image and smoothed with a 3×3 pixel moving boxcar, and then combined (averaged) in a $1^\circ \times 0.5^\circ$ latitude-longitude grid of geographic coordinates. The composite image array is then mapped to an orthographic projection of geographic coordinates centered at 40°N latitude and 12:00 local time (LT) at

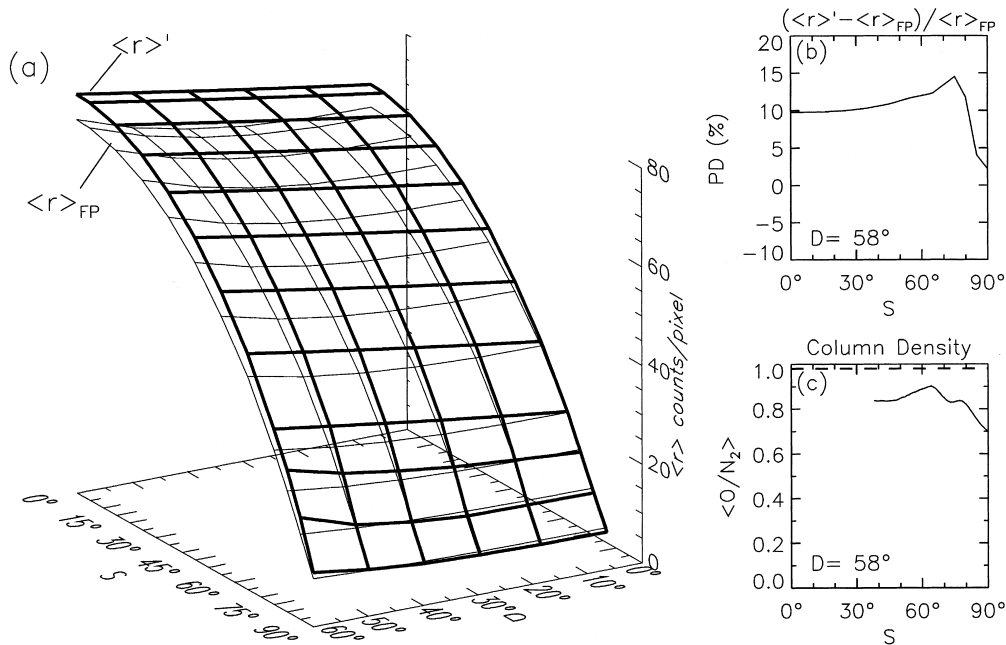


Fig. 5. Comparison of first-principles and empirical modeling results. (a) 3-D surface plots of $\langle r \rangle'$ (from Fig. 3(b), heavy lines) and $\langle r \rangle_{FP}$ (Strickland et al., (1998), light lines) at selected points in the range $0^\circ < S < 90^\circ$ and $0^\circ < D < 90^\circ$. (b) Percentage difference of $\langle r \rangle'$ from $\langle r \rangle_{FP}$ for all S at $D = 58^\circ$. (c) Average MSIS quiet-time column density ratio of O/N_2 ($\langle O/N_2 \rangle$) for the 156 images used to calculate $\langle r \rangle$ at the same values of S and D as the above panel. The solid dashed line indicates $O/N_2 = 0.98$, the value used in the calculation of $\langle r \rangle_{FP}$.

the time of the image obtained midway through the sequence. This is shown in the center panel of Fig. 6, where the colors indicate percentage deviation from quiet-time values as coded in the color bar in the right-hand panel.

The FUV brightness values observed in the composite PD image of Fig. 6 are within $\pm 10\%$ of the quiet-time model except at high latitudes where “contaminating” auroral emissions result in PD values well over 60%. Greater statistical variations are evident at the edge of the composite image close to the terminator, where fewer samples are obtained for calculation of average PD values and counting statistics are more significant. This mapping technique is used in the analysis of the storm-time images of Section 4.

4. Application to storm-time images (days 35–36, 1983)

To demonstrate an application of this refined quiet-time model, we examine images obtained prior to and during a strong geomagnetic storm beginning on day 35 (4 February) of 1983. No solar wind velocity or magnetic field measurements were obtained during this two-day period, precluding a characterization of the solar-wind driver for this storm. However, AE values exceeding 2000 nT at 17:00 and

19:00 UT on day 35 and again at 12:00 UT on day 36 indicate a large input of solar wind energy. The southern hemispheric power index for this period, estimated from DMSF and NOAA satellite passes, exceeded 250 GW in the early hours of the storm, a very high value (Fuller-Rowell and Evans, 1987). The 3-h planetary magnetic index, ap , is around 7 prior to the storm on day 35, rises to 179 at storm onset and reaches 207 at 03:00 UT on day 36.

Images obtained during this storm period reveal dayglow variations at later local times and lower solar zenith angles (S) than were observed in images used to develop the quiet-time model, presenting a test of the $\cos(S)$ -based function used to extrapolate the late-1981 to early-1982 observations to $S = 0^\circ$. An additional test is provided by calculating changes in atmospheric O and N₂ column densities using MSIS (Hedin, 1987) and comparing predicted differences between quiet- and active-time O/N₂ column density ratio and FUV brightness. Strickland et al. (1998) demonstrate the nearly linear dependence of DE-1 photometer response on the ratio of column abundance of O to N₂ in the thermosphere. Column densities are calculated as in the previous section and the Appendix, with a base at the 10^{-5} Torr pressure altitude, corresponding to quiet-time N₂ column densities of approximately 2 to 3 $\times$

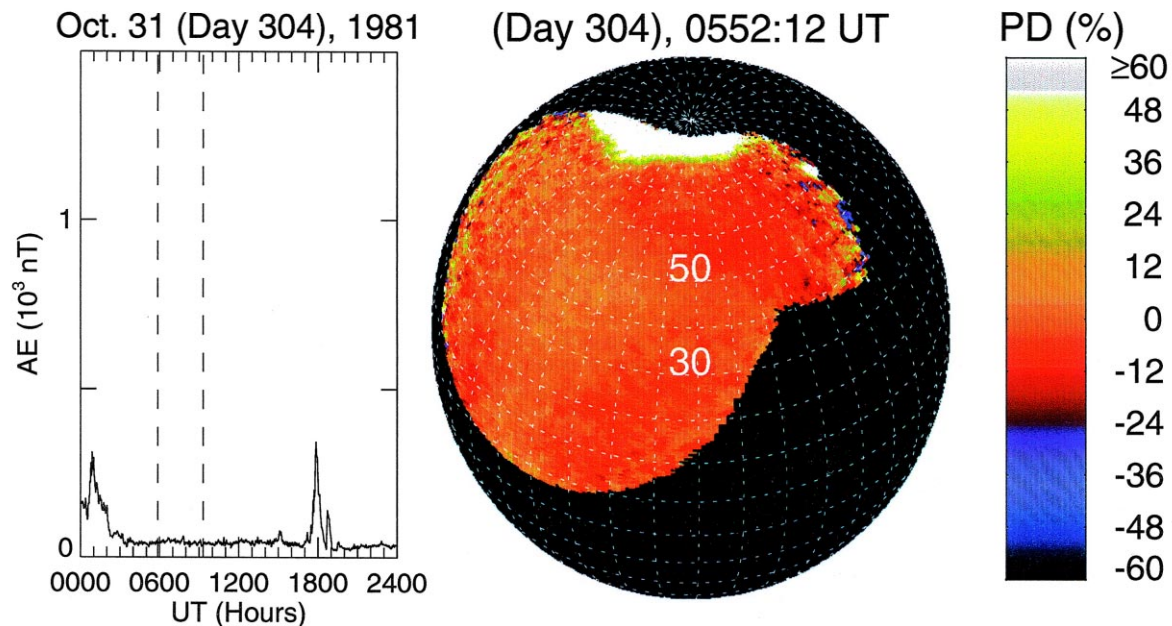


Fig. 6. Composite PD representation of 16 quiet-time images taken by DE 1 on 31 October 1981 between 05:52 and 09:06 UT. AE indices for this day are to the left. The composite PD image is mapped to an orthographic projection of geographic coordinates, with the center of the first projection at 40°N and $\sim 12:00$ LT. The color scale indicates PD values in the $\pm 60\%$ range, as shown in the color bar to the right.

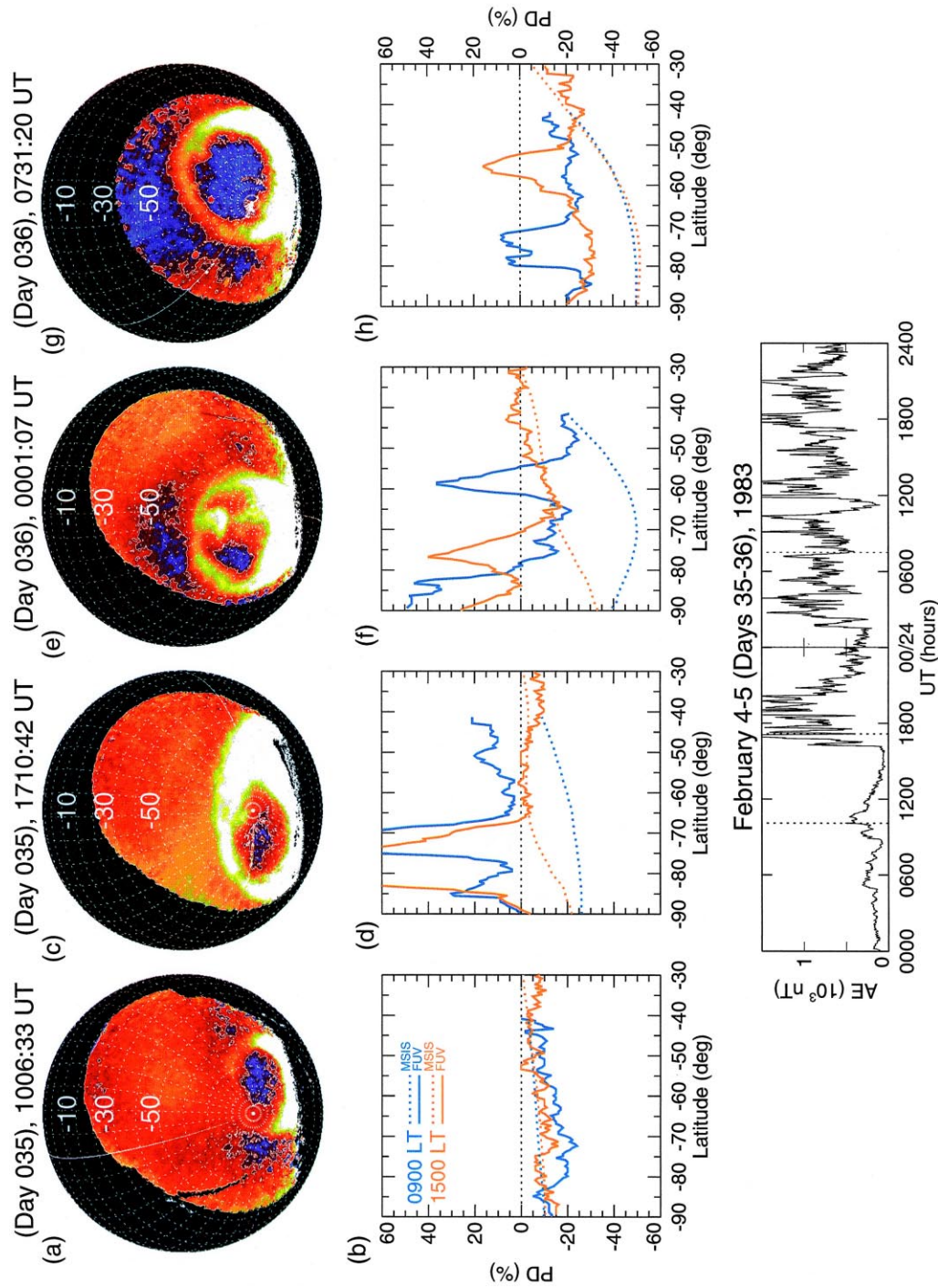


Fig. 7. Series of composite PD images from four successive DE-1 orbits providing coverage of a geomagnetic storm. Images are projected onto an orthographic grid of geographic coordinates, where the center is 60°S and 12:00 LT. (a) PD representations of images obtained between 10:06 and 11:55 UT on February, 4, 1983. (b) Plots of FUV and MSIS PD values at 09:00 and 15:00 LT for the above image. (c,d) Same as image and plot of 7(a) and 7(b), respectively except for 17:10 to 18:59 UT. (e,f) Same as 7(a) and (b), respectively, except for 00:01 to 01:25 UT on 5 February 1983. (g,h) Same as 7(e) and 7(f) except for 07:31 to 08:56 UT.

10^{17} cm^{-2} . All values of a_p , the magnetic index used by MSIS to quantify the level of geomagnetic activity, are set to 2 in the required 60-h history MSIS input for calculation of quiet-time neutral composition. PD values in the column ratio are calculated using these as a reference, just as the dayglow model is used as a reference for calculation of FUV PD values.

Four series of DE-1 FUV images from successive satellite orbits beginning on day 35 are used in the analysis of this storm period. PD values from available images in each orbit are combined (averaged) to form a single PD image. These composite images are presented in Fig. 7 mapped to an orthographic projection of geographic coordinates, and PD values are indicated with the same color bar used in Fig. 6. The satellite orbit plane is at $\sim 14:00$ LT, and motion of the satellite toward apogee over Earth's South Pole results in an elongated imaging signature. The center of each orthographic grid locates 12:00 LT (at the time midway through each imaging sequence) and -60° latitude, with the morning sector to the left. The UT at the beginning of each imaging sequence is indicated above each mapped image, with a vertical dashed line in the AE plot below. The PD values are calculated without corrections for phase angle (P), as the range of phase angles here contains points well outside the range observed in 1981 (there is much more coverage of the PM sector here). The linear fit to $\langle r \rangle$ as a function of P is well behaved in the afternoon sector, but here does not accurately reflect actual variations in O/N_2 across the entire disk. Corrections for F , A , and reduction of instrument sensitivity are otherwise applied to each image before comparison to the values of Fig. 3(b).

The first composite PD image (Fig. 7(a)) shows average PD values from 10 images obtained between 10:06 and 11:55 UT on day 35. These PD values represent slightly disturbed conditions given the values of AE (250–400 nT) and a_p (7) for the period. Decreases below -20% (white contour, blue color) are observed at high latitudes and are most significant in the polar cap. Lesser decreases are observed throughout the imaged area except near 14:00 LT and -60° , where slight increases are observed. The auroral oval is seen at high latitudes near the terminator, where the combination of FUV dayglow and auroral emissions result in PD values over 60%.

The accompanying plot of Fig. 7(b) indicates PD values extracted from the image along the 09:00 and 15:00 LT meridians (shown with solid blue and orange lines, respectively) reflecting the above described variations. Also shown are percent differences in MSIS O/N_2 column density ratios from quiet-time values in these local time sectors (dotted blue and orange lines for 09:00 and 15:00 LT, respectively). MSIS PD values for this image sequence approach -10% at the geographic pole, reflecting the general trend in the FUV

PD values, whose values similarly approach 0% at lower latitudes. Actual a_p values range from 7 to 8 in the required MSIS input for the 60-h period up to the observation time, compared to the value of 2 used to determine the quiet-time values.

For this imaging time, the mid-latitude dayglow signature is quite similar to the quiet-time values reported in Fig. 3(b), even at the sub-solar point. The monotonic decrease in both MSIS and FUV PD values with decreasing latitude is an indication that the dayglow model provides an accurate quiet-time background for this storm period. This is encouraging, since no observations in the range of S seen here were made for the 1981 quiet-time set (e.g., Fig. 3(a)). The $\cos(S)$ -based function provides a realistic modeled quiet-time photometer response for this time of 1983, provided the model correction for the $\sim 50\%$ decrease in photometer sensitivity is applied, as discussed in the Appendix. The comparison of Section 3.1 separately confirmed that the empirical and first-principles modeling results were similar at low S .

The onset of the storm is at $\sim 16:15$ UT, and PD values from 10 images obtained between 17:10 UT and 18:59 UT are shown in Fig. 7(c). PD values from the 09:00 and 15:00 LT meridians are shown as a function of latitude in Fig. 7(d), using the same format as Fig. 7(b). A comparison of this composite image to that obtained in the previous orbit (Fig. 7(a)) shows that, apart from the huge aurora-induced spike in PD values at $< -70^\circ S$, the FUV brightness has increased at nearly all observed local times and latitudes, with the greatest effect in the pre-noon sector. PD values at mid-latitudes which were around -15% are now as high as $+25\%$. The extremely bright and expanded auroral oval is a signature of the high level of activity, and can be seen in the PD image and line plots. Decreases in brightness of -20% are evident in areas of the polar cap not obscured by the immense and asymmetric auroral bulge.

The apparent effect of the sudden onset of magnetic activity is to increase the O/N_2 ratio in the emitting thermospheric column, mainly in the pre-noon sector. While increases in this ratio are expected at low latitudes in the hours after storm onset and in the evening-to-midnight sector of the winter hemisphere (Burns et al., 1995), the increase in brightness at middle latitudes in the summertime morning sector is unexpected and indicates a substantial increase in day-side O/N_2 . A possible explanation of this effect is the transport of nitrogen poor air by meridional neutral winds from high latitudes to dayside middle latitudes, where the winds then dissipate and the transported air downwells, increasing the ratio of O/N_2 . An event as powerful as this may be able to drive portions of the neutral atmosphere to points on the dayside well equatorward of the auroral oval. A strong traveling atmos-

pheric disturbance is another possible influence on the composition and brightness.

In contrast, MSIS indicates significant decreases in O/N_2 column ratios from quiet-time values in the 09:00 LT sector, with differences from quiet-time values approaching -30% at high latitudes (Fig. 7(d), dotted blue line), caused by the increase in ap from 7 to 179. This departs significantly from the increase in O/N_2 ratios inferred from the strong increase in FUV brightness. It is likely that MSIS cannot duplicate the significant departure from diffusive equilibrium possibly caused by the sudden onset of activity. In the 15:00 LT sector, both the FUV and MSIS PD values remain relatively undisturbed (Fig. 3(d), orange solid and dotted lines, respectively), with the exception of the auroral zone where MSIS PD values indicate 10–20% decreases. Comparison with FUV PD values in this area is hampered by the presence of strong auroral emissions which dominate the FUV dayglow.

The composite PD image in Fig. 7(e) from eight observations obtained during the following orbit (between 00:01 and 01:25 UT, day 36) shows that decreases in FUV brightness at mid-latitudes are now present, with strongest effects in the morning sector. In contrast, the afternoon sector has brightened. These effects also are visible in line plots of PD values at 09:00 and 15:00 LT, shown in Fig. 7(f). The significant decreases in PD values in MSIS O/N_2 column densities are also shown. A comparison between PD values reveals a remarkable correspondence between FUV and MSIS measurements equatorward of the auroral oval, the sharply peaked signature of which is apparent in each LT sector of the FUV PD plot. The 09:00 LT sector shows decreases of 20–25% in both FUV brightness and O/N_2 column density ratio. In the afternoon sector, FUV PD values increase from -18 to 0% with increasing latitude, while MSIS values range from -10 to 0% . FUV PD values reach 0% more rapidly, and FUV emissions are brighter equatorward of -50° than in the previous orbit. This is quite possibly a residual effect of the storm-time increase in dayside O/N_2 ratios observed in the previous orbit, a variation which was not accounted for by MSIS.

In the final orbit of the series, decreases in FUV brightness are observed at nearly all points in the DE-1 field of view, as shown in Fig. 7(g). Eight images are included in this final composite PD image from observations between 07:31 and 08:56 UT. The auroral oval is expanded to reveal a large area of decreased brightness in the polar cap, apparently free of discrete auroral features. Plots of MSIS PD values at 09:00 and 15:00 LT in Fig. 7(h) show that MSIS predicts a nearly uniform perturbation in the O/N_2 ratio across the day-side, and an increase in O/N_2 PD values toward the equator. FUV observations within the polar cap at 15:00 LT and at middle latitudes at 09:00 LT demon-

strate a similar trend in FUV PD values south of -50° latitude. Though the trend is similar, FUV PD decreases are only approximately half as great as the MSIS prediction of O/N_2 PD decreases at these latitudes. Equatorward of -50° , afternoon FUV PD decreases are more significant than in the morning. This is in contrast to MSIS O/N_2 PD values which differ little between these sectors. The degree of composition variations observed at particular local times during a complex storm such as that presented here can be expected to depart from empirically determined average values such as those provided by MSIS.

The correspondence of storm time PD values with variations in MSIS column densities is notably good, with some exceptions. The apparent increases in O/N_2 ratios inferred from the increased FUV brightness observed after storm onset are not reflected in MSIS. Conversely, the extremely strong decreases in MSIS O/N_2 at high latitudes after 12 h of intense magnetic activity are not borne out in the FUV observations. From a different modeling perspective, these storm period observations with DE 1 correspond well to first-principles modeling work (e.g. Fuller-Rowell et al., 1996), which demonstrates that storm-time decreases in O/N_2 ratios can be very strong in the noon sector at middle latitudes within 12–15 h of storm onset. The strongest decreases in brightness observed in this imaging series (outside of the polar cap) are in the noon sector 16 hours after initial storm onset. The question of whether a first-principles thermospheric model would duplicate the observation of an increase in O/N_2 at middle latitudes in the morning sector is valid, but is not addressed in this report.

5. Summary

An improved method of calculating reference values for the response of the DE-1 imager to the quiet-time FUV dayglow at wavelengths of 123–165 nm (filter #2) has been developed. The average photometer response $\langle r \rangle$, normalized to $A = 90^\circ$, $P = 90^\circ$, $F = 200$ Jy, is determined, as well as the means to correct any DE-1 image to these fixed values before comparison to the fitted values of the photometer response, $\langle r \rangle'$. The comparison with quiet-time images verifies that the values of $\langle r \rangle'$ offer an accurate representation of the FUV brightness during periods of low activity. Through this empirical modeling of FUV brightness, dayglow variations which correspond well to variations in the quiet-time MSIS model are observed, and a determination of the effective emission altitude is made.

The dayglow reference model described here offers improvements over previous geometric models in the following ways: (1) the empirical model now provides

accurate reference values at $0^\circ < S \leq 105^\circ$ at all $D \leq 65^\circ$, using a fit based on $\cos(S)^n$ and a table of values at $S > 80^\circ$; (2) it provides corrections to FUV dayglow brightness in response to variations in solar EUV and FUV emissions, using F as a proxy for direct solar measurements; (3) corrections are included to simulate variations in photometer response with azimuthal viewing geometry (A), and; (4) local time variations in dayglow brightness are effectively modeled using a new phase angle (P) which, for the first time, allows the empirical model to account for quiet-time neutral composition variations.

Recent work by Strickland et al. (1998) quantitatively compares the empirical quiet-time model by Nicholas et al. (1997) and a first-principles quiet-time model. In that work the difference between the resulting PD values of the two techniques is shown to be significant. The discrepancy can be attributed to the least-squares fit to the quiet-time data used by Nicholas et al. (1997), which diverges from realistic reference values at $S < 40^\circ$ and shows deviations from $\langle r \rangle$ larger than 10% at other ranges of S and D . These issues are addressed in this report and the current empirical model compares very well to first-principles calculations. Strickland et al. and Immel et al. (1997) both present results from Day 267, 1981, where the latter report uses a $\cos(S)^n$ based reference model including azimuth corrections. The anomalous PD values resulting from Strickland et al.'s application of the empirical model are not present in the latter work. Improvements to the empirical model of Nicholas et al. have been demonstrated in earlier work as well (Craven et al., 1994 (Fig. 1)).

The quiet-time reference values provide a baseline FUV brightness for investigation of large-scale variations in dayglow brightness that are observed during periods of strong geomagnetic activity. Section 4 of this work demonstrates that with the refinements to the method detailed in this report, it is possible to identify a variety of thermospheric storm effects due to high latitude inputs using FUV images. The storm of the 4–5 February 1983 study period is monitored during four consecutive DE-1 orbits to reveal the global scale changes in thermospheric composition due to intense high latitude inputs. The level of activity prior to the storm affects the FUV dayglow brightness in a manner consistent with what is expected from MSIS. The development of a large-scale O/N₂ increase at middle latitudes, and its subsequent transport to the afternoon sector, is observed for the first time in DE-1 images. The subsequent negative storm effect is monitored until it eventually dominates the high-latitude thermosphere.

This method will prove useful in future analyses of the dayglow and may be applied to any large set of spectrally separated FUV data for which S , D , A , P

and F are known for each measurement. The technique has been applied to FUV images obtained by the POLAR VIS imager, revealing decreases in OI 130.4-nm brightness in the morning sector at middle latitudes after periods of intense magnetic activity (Craven et al., 1996), a result similar to those from several DE-1 investigations. Further study of variations in the VIS imager reveals, though, that the imager is also sensitive to MUV emissions (200–300 nm) originating much lower in the atmosphere (Frank and Sigwarth, 1999; Immel et al., 1997b). Continued dayglow studies with VIS will require removal of bright cloud-top signatures and the MUV background emissions before geometric modeling methods can be properly applied.

Several instruments now constantly monitor the solar UV spectrum and the solar wind speed, density, and embedded magnetic field. All of these factors have a direct or indirect effect on auroral intensity and thermospheric composition. The current generation of high-altitude spacecraft can monitor Earth's FUV signature on a global scale at higher spatial, temporal and spectral resolution than was available with DE 1, for as many hours as the spacecraft are well away from perigee (e.g., ~12 h for POLAR). Current (POLAR) and future investigations (IMAGE and TIMED) are well suited to the task of generating a large FUV database amenable the type of analysis described in this report for extracting new information about the transfer of energy from the Sun to Earth.

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Appendix A1. Values of coefficients

Coefficients reported for the F and P dependences of $\langle r \rangle$ are for functions fitted to the slopes m_F and m_P at all S . Once fit, the slopes can be expressed as

$$m_F = ((1 - \tan h(S/A - b))/c)^d \quad (A1)$$

and

$$m_P = a \exp(-1*((s - b)/4c)^2) \quad (A2)$$

for the ranges of S described in the respective sections

in this report. For the F dependence, values for a , b , c , and d are 28.9, 3.63, 2.80, and 6.58, respectively. For the P dependence, values for a , b , and c are -0.088 , 62.5 and 9.18 , respectively. No functional form was used for the dependence of $\langle r \rangle$ on A . In this case, the slopes, m , of the individual fits are reported in Table A1 for $S > 81^\circ$ and $D > 20^\circ$, where all values are reported to four digits, significant or not. The influence of A on $\langle r \rangle$ is most significant in this range of S and D .

Coefficients of the fit to $\langle r(A = 90^\circ, P = 90^\circ, F = 200 \text{ Jy}) \rangle$ were reported in Eqs. (2) and (3) of this report. These are used at $S \leq 80^\circ$ and a table of values in $1^\circ \times 1^\circ$ bins of S and D is used for $\langle r \rangle$ at $S > 80^\circ$. A summary of those values shown in Table A2, where the data are reported in larger bins ($\Delta S = 3^\circ$, $\Delta D = 5^\circ$) for convenience.

Corrections to $\langle r \rangle'$ for variations in photometer sensitivity are made for analysis of images obtained in early 1983 using the formula:

$$\langle r(1983) \rangle = \langle r \rangle \cdot c(S, D) \quad (\text{A3})$$

where

$$c(S, D) = 0.436 + D \cdot 0.00556 + S \cdot (0.00159 + D \cdot 0.0000652). \quad (\text{A4})$$

These corrections were determined by least squares fitting the ratio of 1983 quiet-time to 1981 quiet-time photometer response, and values of $c(S, D)$ less than

0.5 are forced to that value. The result is that the slope of the variation with D is zero at $D = 0^\circ$ for all values of S . This requirement accounts for the fact that the variation in contributions of optically thin and thick emissions must identically be zero at the nadir.

Appendix A2. Interpretation of results

The slopes (m) of the variation of $\langle r \rangle$ with F , P , and A , calculated in the iterative process are shown in Figs. A1(a), (b) and (c), respectively. Smoothly varying fits to the values of m are shown for F and P dependencies, and the actual slopes are shown for the A dependence, as no fit was determined. The faint solid lines indicate the values for slopes determined in the initial iteration. The dashed lines indicated the values determined in the second iteration. The final values, used to normalize any DE-1 image to $A = 90^\circ$, $P = 90^\circ$ and $F = 200 \text{ Jy}$, are indicated with heavy solid lines in each plot. The heavy solid line of the third iteration obscures the dashed lines in Figs. A1(a) and (b), as the slopes of the second iteration are very close in value to those of the third.

The slopes shown in Fig. A1(a) clearly show the increase in photometer response with F , reaching ~ 0.1 count/Jansky-pixel at $S = 40^\circ$. The heavy curve is a plot of slopes calculated from Eq. (A1), using the final coefficients reported above. This is consistent with the expected relationship (greater photometer response

Table A1
Slopes of linear fit to $\langle r \rangle$ vs A^a

$D \downarrow S \rightarrow$	81–84°	84–87°	87–90°	90–93°	93–96°	96–99°	99–102°	102–105°
20–25°	0.003 ± 0.003	0.005 ± 0.002	0.002 ± 0.003	0.006 ± 0.002	0.006 ± 0.001	0.003 ± 0.001	0.002 ± 0.001	0.000 ± 0.002
25–30°	0.004 ± 0.019	-0.005 ± 0.009	0.004 ± 0.003	0.006 ± 0.003	0.0059 ± 0.0012	0.006 ± 0.002	0.002 ± 0.001	0.0010 ± 0.0004
30–35°	-0.006 ± 0.012	-0.002 ± 0.010	0.013 ± 0.003	0.017 ± 0.004	0.017 ± 0.004	0.007 ± 0.001	0.001 ± 0.001	0.0010 ± 0.0003
35–40°	-0.002 ± 0.016	0.015 ± 0.008	0.029 ± 0.005	0.026 ± 0.003	0.014 ± 0.003	0.008 ± 0.001	0.002 ± 0.001	0.008 ± 0.001
40–45°	0.014 ± 0.010	0.030 ± 0.012	0.027 ± 0.003	0.020 ± 0.005	0.017 ± 0.002	0.010 ± 0.001	0.003 ± 0.002	0.0011 ± 0.0003
45–50°	0.033 ± 0.017	0.025 ± 0.005	0.028 ± 0.007	0.023 ± 0.003	0.024 ± 0.002	0.013 ± 0.001	0.005 ± 0.002	0.001 ± 0.001
50–55°	0.025 ± 0.006	0.032 ± 0.003	0.043 ± 0.008	0.036 ± 0.002	0.026 ± 0.002	0.019 ± 0.003	0.006 ± 0.001	0.003 ± 0.001
55–60	0.038 ± 0.007	0.043 ± 0.009	0.047 ± 0.012	0.046 ± 0.004	0.037 ± 0.003	0.027 ± 0.004	0.010 ± 0.002	0.0004 ± 0.0012
60–66°	0.044 ± 0.027	0.067 ± 0.001	0.079 ± 0.015	0.073 ± 0.017	0.090 ± 0.019	0.043 ± 0.014	0.023 ± 0.005	0.001 ± 0.001

^a Values of the slope of linear variation of $\langle r \rangle$ with A in bins $\Delta S = 3^\circ$, $\Delta D = 5^\circ$, with uncertainty in the slope reported.

Table A2

 $\langle r(A = 90^\circ, P = 90^\circ, F = 200 \text{ Jy}) \rangle$ in bins of S and D^a

$D \rightarrow$	0°	5°	10°	15°	20°	25°	30°	35°	40°	45°	50°	55°	60°	(σ_m)
78°	23.55	23.64	22.70	22.93	23.18	23.49	23.56	23.76	23.87	25.19	26.45	27.65	32.25	0.40
81°	18.71	18.62	18.38	18.64	18.56	19.03	19.10	18.81	19.73	20.58	21.30	21.96	26.99	0.34
84°	14.17	14.29	14.43	14.58	14.70	14.40	14.38	14.63	15.52	15.66	16.38	16.91	21.97	0.32
87°	9.85	10.38	10.50	10.46	10.40	10.23	10.32	10.99	11.10	11.34	11.85	12.44	14.43	0.24
90°	6.88	6.63	6.66	6.67	6.72	6.75	7.08	7.19	7.31	7.38	7.87	8.13	9.88	0.26
93°	4.12	3.84	3.82	3.85	3.88	3.97	4.13	4.11	4.19	4.26	4.35	4.74	5.12	0.19
96°	1.83	1.87	1.90	1.97	1.91	2.05	1.99	1.95	1.99	2.01	2.07	2.07	2.29	0.13
99°	0.81	0.80	0.94	0.96	0.97	0.95	0.92	0.89	0.89	0.84	0.86	0.92	1.03	0.14
102°	0.53	0.51	0.59	0.59	0.58	0.57	0.53	0.53	0.49	0.46	0.44	0.46	0.56	0.08
105°	0.47	0.41	0.43	0.42	0.47	0.45	0.41	0.39	0.37	0.37	0.35	0.36	0.46	0.06
108°	0.41	0.37	0.38	0.43	0.46	0.41	0.39	0.35	0.38	0.37	0.38	0.38	0.91	0.07

^a Smoothed values of $\langle r \rangle$ in bins $\Delta S = 3^\circ$, $\Delta D = 5^\circ$. Average uncertainties in the mean values for each bin of S are reported in the rightmost column. Reported values are equivalent to $\langle r \rangle'$ in this range.

with greater solar output), though the assumption of a linear relationship between OI brightness and F over the entire quiet-time range of $127 \text{ Jy} < F < 296 \text{ Jy}$ is an approximation which appears to be justified by forward modeling work of Strickland et al., (1995).

The variation in $\langle r \rangle$ with P is marked by mostly negative slopes as noted in Fig. A1(b). The heavy solid line is calculated using Eq. (A2) and the coefficients reported above. This indicates that the OI FUV brightness diminishes as observations tend toward earlier local times (increasing P). This corresponds well to the variation in relative column densities between O and N_2 as indicated by MSIS (Hedin, 1987). To compare photometer response to column densities, it is important to select an altitude of the base of the column based on the local composition (vs a single fixed altitude). Strickland et al. (1998) select the altitude corresponding to an N_2 column density of 10^{17} cm^{-2} . In this report we use instead the altitude at which the atmospheric pressure is equal to 10^{-5} Torr, typically between 125–135 km and changing with solar activity. The resulting reference column density of N_2 is usually in the range of $2\text{--}3 \times 10^{17} \text{ cm}^{-2}$.

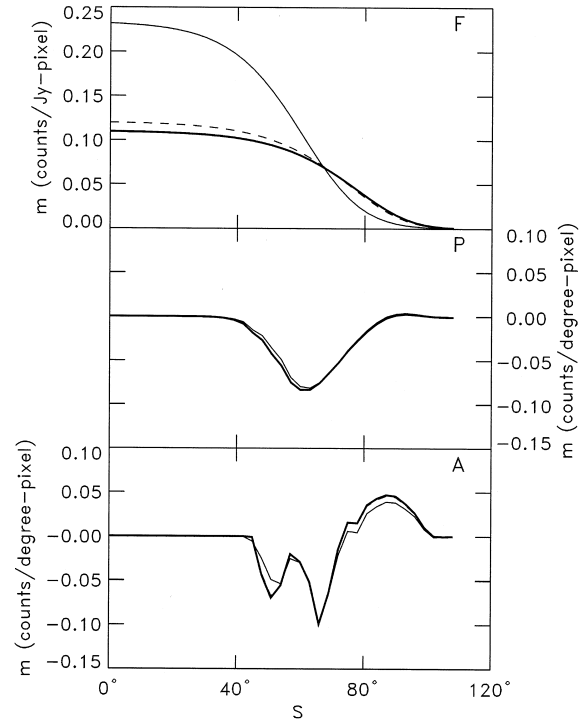


Fig. A1. Iteratively determined fits to slopes of $\langle r \rangle$ vs F and P and iteratively determined slopes of $\langle r \rangle$ vs A at all S . In all cases the heavy line indicates final values. (a) Fit to slope of $\langle r \rangle$ vs F . (b) Fit to slope of $\langle r \rangle$ vs P . (c) Slope of $\langle r \rangle$ vs A at $50^\circ \leq D < 55^\circ$ and three increasing ranges of S .

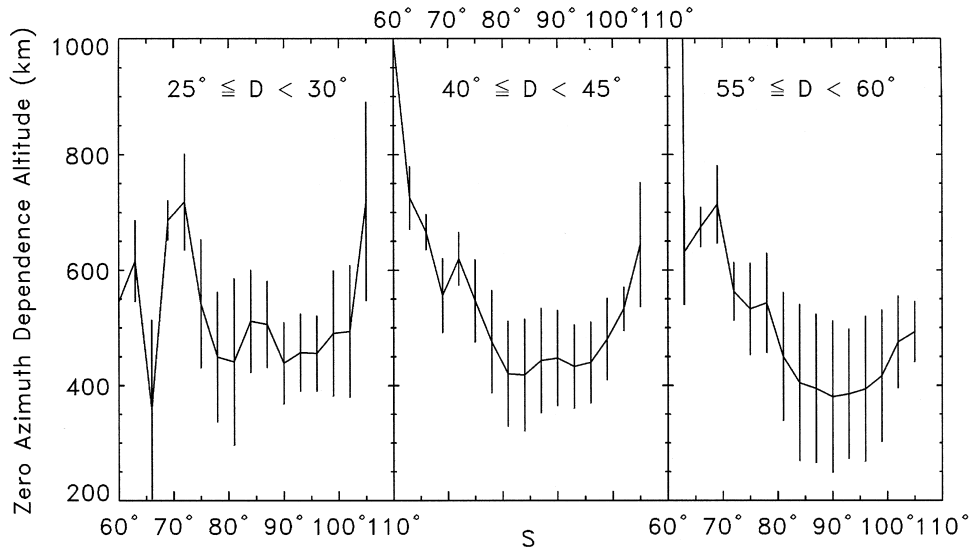


Fig. A2. Effective emission altitude calculated for $60^\circ \leq S < 105^\circ$ in bins of D ($\Delta D = 5^\circ$) at (a) $D = 25^\circ$, (b) $D = 40^\circ$ and (c) $D = 55^\circ$. Vertical bars at each point indicate uncertainties.

The MSIS atmosphere is calculated for each of the pixel locations in the quiet-time images and average column densities of $\langle O \rangle$ and $\langle N_2 \rangle$ are determined in the same manner as $\langle r \rangle$. One can then determine the correlation coefficient between $\langle r \rangle$ and $\langle O \rangle$ or $\langle N_2 \rangle$ for each range of S . The average correlation coefficient (for $48^\circ < S < 105^\circ$) between $\langle r \rangle$ and $\langle O \rangle$ is 0.71. The average correlation coefficient between $\langle r \rangle$ and $\langle N_2 \rangle$ is -0.63 . Lower values of S are excluded due to the high uncertainties in the values of m . Thus we are confident that the dependence on P is rooted in an actual quiet-time variation of O/N_2 in the thermosphere.

The effect of azimuth angle, A , on photometer response originates in the selection of an assumed emission altitude, so chosen even though it is certain that there is no single OI 130.4-nm emission altitude. Furthermore, the 130.4-nm source function changes with S . With the selection of a single emission altitude, there will be a variation in observed brightness with the azimuth of observation at some range of S . The effect will be null at low values of D but significant at $D > 50^\circ$. The slopes shown in Fig. A1(c) indicate that the selection of 500 km for this research is too low at $S < 72^\circ$ and too high closer to the terminator. The slopes in the higher ranges of S and D are most significant in correcting the observed photometer response. These slopes and their uncertainties for each $3^\circ \times 5^\circ$ bin of S and D are given in Table A1.

By determining a linear fit to $\langle r \rangle$ as a function of A at all S and D at a range of altitudes, one can find the assumed emission altitude which eliminates the variation with A . This is shown in Fig. A2 for three ranges

of D at $60^\circ < S < 110^\circ$. The 500-km altitude, selected for this research, is within the range of error for many of these determinations of effective emission altitude. This is surprisingly high, as the OI 130.4-nm source function undoubtedly has a maximum below 300 km (away from the terminator) (Meier, 1991; Strickland and Thomas, 1976). The assumption of a linear relationship between $\langle r \rangle$ and A is likely an oversimplification and may be the source of this discrepancy. However, a higher order fit is not useful, given that observations in many S and D ranges are limited to only three or four 30° bins in A (e.g., Fig. 2(c)).

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